# Support Vector Machines: Primal, Dual, and Kernelized Formulations

## 1 Support Vector Machines (SVM)

Support Vector Machines (SVMs) are supervised learning methods for classification (and regression) that construct a decision boundary with the largest possible margin between classes. The kernelized version allows nonlinear decision boundaries using the kernel trick.

### 1. Linear Classifiers and Motivation

We consider binary classification with training data

$$(x_i, y_i), \quad i = 1, \dots, N, \quad x_i \in \mathbb{R}^d,\ y_i \in \{-1, +1\}.$$

A linear classifier predicts

$$f(x) = \operatorname{sign}(w^\top x + b), \qquad w \in \mathbb{R}^d,\ b \in \mathbb{R}.$$

The challenge: many separating hyperplanes may exist. Which one should we choose?

### 2. Geometric Intuition: The Margin

- The *margin* of a classifier is the distance between the decision boundary and the closest data points.
- Maximizing the margin leads to better generalization (a large margin classifier is less sensitive to small perturbations of the data).

For a hyperplane $(w, b)$, the functional margin of $(x_i, y_i)$ is

$$\gamma_i = y_i(w^\top x_i + b).$$

The geometric margin is obtained by normalizing by $\|w\|$:

$$\hat{\gamma}_i = \frac{y_i(w^\top x_i + b)}{\|w\|}.$$

### 3. Hard-Margin SVM

If the data are linearly separable, the hard-margin SVM solves:

$$\min_{w,b}\ \tfrac{1}{2}\|w\|^2 \quad \text{subject to } y_i(w^\top x_i + b) \geq 1,\ \ i = 1, \dots, N.$$

- The objective $\frac{1}{2}\|w\|^2$ maximizes the margin.
- The constraints ensure correct classification with margin at least 1.

## 4. Soft-Margin SVM

For non-separable data, introduce slack variables $\xi_i \geq 0$:

$$\min_{w,b,\xi} \tfrac{1}{2}\|w\|^2 + C\sum_{i=1}^{N}\xi_i \quad \text{subject to} \;\; y_i(w^\top x_i + b) \geq 1 - \xi_i.$$

- Parameter $C > 0$ controls the trade-off between margin size and misclassification penalty.
- Large $C$: fewer margin violations (risk of overfitting).
- Small $C$: wider margin, more tolerance to misclassification (risk of underfitting).

## 5. Dual Problem and KKT Conditions

The primal soft-margin problem can be expressed as a Lagrangian and solved in dual form. The dual optimization problem is:

$$\max_{\alpha\in\mathbb{R}^N} \sum_{i=1}^{N}\alpha_i - \tfrac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\alpha_i\alpha_j y_i y_j (x_i^\top x_j),$$

subject to

$$\sum_{i=1}^{N}\alpha_i y_i = 0, \qquad 0 \leq \alpha_i \leq C.$$

- The solution involves only inner products $x_i^\top x_j$.
- The classifier is
$$f(x) = \operatorname{sign}\Big(\sum_{i=1}^{N}\alpha_i y_i (x_i^\top x) + b\Big).$$
- The Karush-Kuhn-Tucker (KKT) conditions characterize optimality:
$$\alpha_i\big(y_i(w^\top x_i + b) - 1 + \xi_i\big) = 0, \quad (C - \alpha_i)\xi_i = 0.$$
These imply that only support vectors ($\alpha_i > 0$) lie on or inside the margin.

## 6. Connection to Hinge Loss

The primal soft-margin SVM is equivalent to minimizing the empirical hinge loss with $\ell_2$ regularization:

$$\min_{w,b} \sum_{i=1}^{N}\max\big(0, 1 - y_i(w^\top x_i + b)\big) + \tfrac{\lambda}{2}\|w\|^2.$$

## 7. Support Vectors

- Points with $\alpha_i > 0$ are called *support vectors*.
- They are the critical points that define the decision boundary.
- Non-support vectors ($\alpha_i = 0$) do not influence the classifier.

Thus, the SVM solution is typically sparse: the boundary depends only on a subset of the data.

## 8. Kernel SVM

To handle nonlinear separations, map inputs into a Hilbert space $H$ with $\varphi : X \to H$. The dual problem becomes

$$\max_{\alpha \in \mathbb{R}^N} \sum_{i=1}^{N} \alpha_i - \tfrac{1}{2} \sum_{i,j=1}^{N} \alpha_i \alpha_j y_i y_j \, \kappa(x_i, x_j),$$

subject to

$$\sum_{i=1}^{N} \alpha_i y_i = 0, \qquad 0 \le \alpha_i \le C,$$

where $\kappa(x, z) = \langle \varphi(x), \varphi(z) \rangle_H$ is a kernel.

The decision function is

$$f(x) = \operatorname{sign}\Big( \sum_{i=1}^{N} \alpha_i y_i \kappa(x_i, x) + b \Big).$$

- Common kernels: linear, polynomial, Gaussian RBF, sigmoid.
- Kernel trick: computation uses only $\kappa(x_i, x_j)$, not explicit $\varphi(x)$.

## 9. Practical Remarks

- SVM is convex: the optimization has a unique global optimum.
- Scaling of data strongly influences performance (standardization recommended).
- Choice of kernel and parameters ($C$, kernel parameters like $\sigma$ in RBF) is typically done via cross-validation.
- SVMs are sparse (depend only on support vectors), making them efficient at prediction once trained.

## 10. Summary

- Hard-margin SVM: maximum-margin classifier for separable data.
- Soft-margin SVM: adds slack variables to handle overlap/noise.
- Dual SVM: involves only inner products and support vectors.
- Kernel SVM: replaces inner products with kernel evaluations, enabling nonlinear classification.
- Equivalent to hinge-loss minimization with regularization.